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Emosense: Multi Emotional System Using Deep Learning for Customer Feedback

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Project Abstract

EmoSense is a cutting-edge system that analyzes customer emotions in product review videos from platforms like YouTube, TikTok, and Instagram. Using the Multimodal EmotionLines Dataset (MELD), which includes over 13,000 labeled utterances from the TV series *Friends*, the system combines text, audio, and visual data to identify emotions (e.g., happiness, anger, sadness) and sentiments (positive, negative, neutral). Built with Python and PyTorch, EmoSense offers a simple mobile app for real-time emotion analysis and tracking over time, helping businesses gain valuable customer insights. The system achieves a test accuracy of 67.62% and an F1-score of 66.64%, performing well on neutral emotions but struggling with less common emotions like disgust and anger due to dataset imbalances. Its reliance on scripted dialogue limits performance on unscripted reviews.

Keywords: Multimodal Emotion Recognition, Affective Computing, MELD Dataset, Customer Sentiment Analysis, Social Media Reviews, Video Analysis, Machine Learning, Deep Learning, Transformer Models, PyTorch, Human-Computer Interaction, Product Recommendation

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Chapter 1

Introduction

1.1 Background

Emotion recognition from multimedia content is a critical area in affective computing, enabling applications in human-computer interaction, market research, and customer service. Videos offer a rich source of multimodal data, including text (from subtitles or transcripts), audio (tone and pitch), and visual cues (facial expressions and body language), which can be analyzed to detect human emotions. This capability is particularly valuable for understanding customer sentiments expressed in product reviews on social media platforms.

Platforms like YouTube, TikTok, and Instagram host millions of user-generated videos, many of which are reviews of products such as food, technology, or consumer goods. These reviews often convey emotions that reflect customer satisfaction, preferences, or dissatisfaction. However, manually analyzing such content is time-consuming and impractical due to its volume and diversity. Automated systems that can process and interpret emotions from videos are thus essential for businesses seeking to gain insights into customer perceptions. The Multimodal EmotionLines Dataset (MELD) (MELD Dataset), which includes annotated dialogues from the TV series *Friends*, provides a robust foundation for developing such systems by offering labeled multimodal data for emotion recognition.

1.2 Motivation

The motivation for this project lies in the growing need for businesses to understand customer emotions in the digital era. Video reviews on social media platforms capture nuanced emotional responses that traditional feedback methods, such as surveys or written reviews, may overlook. For instance, a customer's tone of voice or facial expression in a tech product review can reveal excitement or frustration, providing deeper insights than text alone. By automating the analysis of these videos, businesses can quickly identify trends, address customer concerns, and improve their offerings.

The proliferation of social media as a platform for consumer expression further underscores the need for scalable emotion recognition tools. An automated system can process large volumes of video content, enabling businesses to stay competitive by responding promptly to customer feedback. This project aims to bridge the gap between affective computing research and practical customer service applications.

1.3 Objectives

The primary objective of this project is to develop a multimodal emotion recognition system that analyzes product review videos from social media platforms. The specific goals are:

- To process videos uploaded from platforms such as YouTube, TikTok, and Instagram.
- To extract and classify emotions (e.g., happiness, sadness, anger, surprise) and sentiments (positive, negative, neutral) using text, audio, and visual data.
- To provide businesses with actionable insights based on the detected emotions, enhancing customer service and product development.

The project initially leverages the MELD dataset to train and evaluate the emotion recognition model. The techniques developed will be adapted to handle unscripted, user-generated review videos, addressing the unique challenges of social media content.

1.4 Scope

The project focuses on the following areas:

- **Data Sources**: Videos from YouTube, TikTok, Instagram, and similar platforms, specifically product reviews of food, technology, and other consumer goods.
- **Modalities**: Text (subtitles or transcripts), audio (voice features), and visual (facial expressions and gestures) data extracted from videos.
- **Emotions and Sentiments**: Classification into standard emotion categories (e.g., anger, joy, neutral) and sentiment labels (positive, negative, neutral).

- **Dataset**: The MELD dataset, containing over 1,400 dialogues and 13,000 utterances with multimodal annotations, is used for model development and testing. Future work may involve creating a custom dataset of review videos.

The project does not cover real-time video analysis or non-review content, such as tutorials or vlogs, unless they contain explicit product evaluations.

1.5 Significance of the Study

This project advances affective computing by applying emotion recognition to a real-world problem: understanding customer emotions in social media reviews. By utilizing the MELD dataset, a well-established resource in the research community, the project builds a robust model that can potentially generalize to diverse video types. The resulting system offers businesses a practical tool to analyze customer feedback at scale, leading to improved decision-making and customer satisfaction.

Additionally, the project explores the transferability of models trained on scripted, conversational data (like MELD) to unscripted, user-generated content. This contributes to the broader field of multimodal learning, demonstrating how techniques developed for one domain can be adapted to another, paving the way for more versatile emotion recognition systems.

1.6 Outline of the Report

This report is organized as follows:

- **Chapter 2: Related Work** - Reviews literature on emotion recognition, multimodal learning, and applications to social media content.
- **Chapter 3: Materials and Methods** - Describes the system architecture, model design, MELD dataset, and development methodology.
- **Chapter 4: Implementation and Results** - Details the implementation, experiments conducted with the MELD dataset, and evaluation results.
- **Chapter 5: Discussion** - Interprets the results, compares findings with prior studies, and discusses limitations and challenges.
- **Chapter 6: Conclusion and Future Work** - Summarizes the project outcomes and outlines future directions, including adaptation to review videos.

Chapter 2

Related Work

2.1 Introduction to Literature Review

This chapter provides a comprehensive review of the literature on multimodal emotion recognition in conversations (MERC), with particular focus on the MELD dataset and the challenges of adapting such models to real-world applications like customer review feedback analysis. The field of multimodal emotion recognition in conversations (MERC) has experienced significant advances since the introduction of comprehensive, aligned datasets such as MELD. Research has expanded from unimodal text-based techniques to sophisticated deep learning methods leveraging text, audio, and visual modalities. Within this context, recent work has emphasized improved fusion methodologies, context modeling, and evaluation protocols, particularly on the MELD dataset, to better manage the inherent challenges of emotion recognition—especially class imbalance and the complexity of multimodal fusion.

However, despite these advances, key challenges remain, including standardized reporting practices (e.g., per-class F1), robust handling of minority emotions, and the transfer of learned representations to distinct domains such as customer review feedback.

2.2 Historical Perspective

The trajectory of research on multimodal emotion recognition in conversation (MERC) can be traced through distinct developmental stages, each marked by the introduction of new datasets, methodological paradigms, and theoretical refinements. Central to this progression is the MELD dataset, which catalyzed much of the recent work by providing a large-scale, finegrained, multimodal resource tailored for emotion analysis in complex, multi-party dialogues [13].

2.2.1 Dataset Foundations and Early Baselines (2018–2019)

The publication of MELD [19] addressed a critical gap by delivering synchronized text, audio, and visual tracks, with consistent annotation across seven emotion classes. Its design enabled the evaluation of models that could exploit not just linguistic content but also paralinguistic cues (prosody, facial expression) and conversational context. Initial studies primarily established unimodal and simple early-fusion baselines, demonstrating that while text alone provides strong performance, fusing additional modalities could theoretically enhance emotion detection [13, 19].

2.2.2 First Wave of Deep Multimodal Models

With the growing adoption of pre-trained language models such as BERT and the rise of deep learning, researchers began to integrate advanced textual encoders alongside increasingly sophisticated audio and visual representations [14, 5, 16]. Early works experimented with concatenation-based fusion, but these approaches often fell short in fully exploiting higher order inter-modal relationships. Attention mechanisms and transformer-based fusion later became standard, yielding improved aggregate metrics [19, 14, 16].

2.2.3 Emergence of Contextual and Graph-Based Modeling

Recognizing the importance of dialogue context and speaker interplay for emotion understanding the field progressed towards context-aware modeling. Sequential models (e.g., RNN, BiLSTM) incorporated information from neighboring utterances, but it was the adaptation of graph neural networks (GNNs)—explicitly modeling conversation as a set of interconnected utterances and speakers—that marked a key advance [9, 12, 7]. This move enabled structured reasoning across dialogue structure and speaker roles, enhancing the model's capacity to disambiguate subtle emotional cues arising from conversational interplay [9, 7].

2.2.4 Advanced Fusion and Architectural Diversification (2021–2023)

A parallel trend developed toward more nuanced fusion strategies. Cross-modal attention, hierarchical transformers, and novel staged or two-stage frameworks allowed for dynamic weighting of modalities and information pathways [3, 19, 14, 16]. Embrace Net-style architectures further enhanced robustness to missing or noisy modalities [19]. The introduction of hybrid fusion protocols—combining feature-level, model-level, and decision-level integration—improved the system's ability to capture fine-grained modal interactions [18].

2.2.5 Imbalanced Learning, Contrastive Approaches, and Evaluation Reform

MELD's pronounced class imbalance, with minority emotions often comprising only a small Proportion of the data motivated the field to adopt targeted countermeasures [12, 14]. These include generative adversarial data augmentation [12], margin-based and focal losses [12, 14], and graph contrastive learning [15, 1], all designed to sharpen class boundaries and retain sensitivity to underrepresented emotions. Simultaneously, strong advocacy emerged for reporting metrics (macro-F1, class-level F1) that illuminate minority-class performance, though the latter remains inconsistently provided [12, 14].

2.2.6 Unified Benchmarks and Reproducibility Drives

Recent meta-research has underscored the lack of standardized evaluation protocols and Reproducibility as a bottleneck for progress [10]. Efforts like MER Bench aim to harmonize feature extraction, fusion, and evaluation, serving as community resources for apples-to apples comparison and signaling the field's maturation [10].

2.2.7 Limitations and Unmet Needs

Despite these methodological advances, notable limitations persist. Many top-line results continue to report only aggregate F1 or accuracy without full class-level breakdowns, obscuring true progress for minority emotions [17, 6, 8, 5, 16]. Furthermore, while the literature recognizes the high value of multimodal fusion on MELD, many" tri-modal" papers actually present bimodal results due to practical challenges with visual data [17, 20]. Perhaps most importantly, no existing work directly applies or adapts MELD-based multimodal recognition to the domain of customer review feedback—an open question with real-world significance [10].

2.3 Theoretical Framework

Multimodal emotion recognition in conversations (MERC) rests on an evolving body of theory that synthesizes perspectives from machine learning, signal processing, natural language understanding, and graph-based modeling. The modern framework for MERC—particularly as applied to the MELD dataset—encapsulates several interlocking components that underpin recent methodological advances.

2.3.1 Multimodal Representation Learning

At the core, MERC assumes that emotional states are encoded in multiple, partially redundant modalities—primarily text, audio, and visual cues [13, 14, 5, 16]. The theoretical challenge is to:

- Extract salient, discriminative features from each modality
- Align heterogeneous feature spaces across modalities, and
- Fuse these features to produce a joint representation that supports accurate emotion Classification.

Modern approaches draw upon deep learning for modality-specific encoding:

- Text: Pretrained language models (notably BERT and its variants) produce contextually rich, token-level embeddings, capturing subtle semantic and emotional nuances [14, 5,16].
- Audio: Either handcrafted features (e.g., MFCCs), or deep audio embeddings (e.g., from wav2vec2.0 or learned CNNs) capture prosodic, spectral, and paralinguistic cues relevant for affect [14, 5].
- Visual: Deep CNNs (e.g., ResNet-50) or 3D CNNs extract expressive facial features, micro-expressions, and gesture cues from face tracks or frames [14, 5, 16].

2.3.2 Multimodal Fusion Mechanisms

The principal theoretical problem in MERC lies in fusing psychologically and statistically divergent cues from different modalities, each of which may be only partially informative in a given context [14, 18]. Fusion is classically categorized as:

- Early Fusion (feature-level): Direct concatenation of learned embeddings from each modality, followed by a shared deep classifier [5].
- Late Fusion (score-level): Independent modeling per modality, combined through weighted voting or shallow merging at the output stage.
- Hybrid/Attention-based Fusion: Theoretically favored in recent work, these methods employ cross-modal transformers, attention mechanisms, or dynamic fusion gates to model interdependencies and adaptively fuse representations in a context-sensitive manners [3, 19, 14, 16, 18].

This enables the model to prioritize the most informative modalities at each stage, leveraging both intra- and inter-modal cues. More recent innovations introduce graph-based fusion, were utterances and modalities are represented as nodes and edges in a graph structure, with GNNs or related architectures capturing relationships across both conversation and modalities [12, 9, 15, 1, 2, 7].

2.3.3 Contextual and Sequential Modeling

Emotional meaning in conversations is inherently contextual and sequential: the affective state of an utterance is shaped by preceding and following turns, speaker identity, and dialog flow. The theoretical framework for modeling this context has shifted from simple RNN/BiLSTM architectures—which capture short-term dependencies—to more expressive graph-based or hierarchical transformer models that encode both speaker roles and long-range dependencies across conversational utterances [12, 9, 15, 1, 19, 7].

Such models make use of:

- Utterance graphs: Nodes represent utterances, connected based on time, speaker, or other logical flows [9, 2, 7].
- Speaker embeddings or identity signals, which distinguish emotion trajectories across different conversational participants [12].

2.3.4 Imbalanced Learning and Loss Design

MELD and similar datasets are characterized by severe class imbalance, with emotions like” fear” and ”disgust” are underrepresented. Directly optimizing for accuracy or na¨ıve crossentropy can result in models that ignore these minority classes, thus failing to serve the full spectrum of effective outcomes. The theoretical framework increasingly incorporates:

- Cost-sensitive learning and loss reweighting: To penalize misclassification of rare classes more heavily.
- Margin-based, focal, or adversarial losses: Explicitly designed to encourage clearer separation of minority class boundaries and counteract over-representation of majority classes [12, 15, 14].
- Generative data augmentation: Using, for example, multimodal adversarial networks (GANs) to synthesize minority-class training examples [12].

2.3.5 Contrastive and Graph Representation Learning

Graph contrastive learning (GCL) extends the theoretical toolkit by maximizing the discriminative distance between instances of different emotion classes while minimizing intra-class distances, within and across modalities [15, 1]. Such approaches augment the feature and fusion learning process by explicitly encoding both similarity (for the same class) and dissimilarity (for different classes), leading to sharper class boundaries, especially crucial in imbalanced settings.

2.3.6 Evaluation Theory

The field's consensus is that macro-F1 should serve as the principal metric due to the set-up's inherent imbalance [12, 14]. The full theoretical recommendation is to report both aggregate macro-F1 and class-level F1 (per emotional category), as only the latter reveals true model robustness and progress on rare emotions [12].

2.3.7 Domain Transfer and Adaptation (Unrealized Theory)

While cross-domain and transfer learning methods are theoretically foregrounded as a necessity for real-world deployment—especially for adaptation from highly-scripted conversational data (MELD) to domains like customer review feedback—no empirical studies in the current literature demonstrate this transfer concretely [10]. The theoretical underpinnings suggest a need for domain alignment mechanisms, adversarial adaptation, and few-shot learning to bridge the substantial linguistic and distributional gap between source and target domains.

2.4 Previous Research and Studies

Research on multimodal emotion recognition in conversations (MERC)—specifically on the MELD dataset—has evolved rapidly, reflecting broader trends in natural language processing, speech analysis, computer vision, and deep learning. This review synthesizes major methodological developments, representative works, and persistent gaps, with close attention to the nuances of dataset properties, challenges in multimodal fusion, architectural advances, class imbalance, evaluation, and limitations—especially regarding transfer to customer review domains.

2.4.1 Foundational Efforts: Data and Early Baselines

The pivotal event in this subfield was the release of the MELD dataset [13]. MELD provided, for the first time, a large-scale, English-language, multi-party conversational corpus with carefully aligned text, audio, and visual data per utterance. Crucially, it supplied gold standard annotations for seven emotion categories—neutral, joy, sadness, anger, surprise, fear, and disgust—which established a rigorous foundation for tri-modal emotion research. Early studies focused on establishing performance baselines. Poria et al. [13] demonstrated the benefit of context modeling—representing dialogue not just as isolated utterances but as sequences with inter-speaker dependencies—via both unimodal and simple concatenation-based multimodal models. They showed that, although text encoded the most informative cues, audio and vision each offered modest but real improvements in accuracy and F1 when added to the fusion pipeline. However, these early works used only shallow neural architectures.

2.4.2 Move to Deep Learning and Feature-Rich Architectures

With the mainstreaming of deep learning, the field shifted from shallow models and hand engineered features to representation learning pipelines based on modern encoders:

- Text: The adoption of the BERT architecture and its variants allowed for contextually rich, semantic embeddings capturing fine-grained affective subtext [14, 5, 16]. BERT6 based models outperformed earlier LSTM or RNN approaches, showing that access to pre-trained, transformer-based language understanding was a critical advance.

- Audio: Researchers experimented with MFCC features, deep CNNs for spectrogram processing, and, more recently, representations from self-supervised models such as wav2vec2.0 [14, 5].

- Vision: Deep CNNs, notably ResNet-50 and its variants, were widely used to extract visual features from face crops or image sequences [14, 5, 16]. Some studies explored 3D CNNs or temporal pooling to integrate dynamic facial expressions over short windows.

Simple feature concatenation became the basic" early fusion," serving as a reference for More advanced approaches [5]. Although such models typically surpassed unimodal systems, their inability to model modality-specific uncertainties or inter-modal relationships limited their effectiveness on MELD's noisy, imbalanced, multiparty data.

2.4.3 Rise of Attention Mechanisms, Transformers, and Advanced Fusion

A major methodological turning point occurred with the integration of attention-based fusion and transformer architectures. These models, motivated by achievements in adjacent NLP and vision fields, enabled the system to learn where to focus—dynamically prioritizing cues from each modality and adaptively weighting their influence on final predictions [3, 19, 14, 16, 18].

- Transformer Fusion & EmbraceNet (Xie et al. [19]): Employed a transformer based, cross-modal attention mechanism within the EmbraceNet framework, showing that dynamic fusion not only raised aggregate accuracy but also conferred a degree of robustness to missing or corrupted modalities. This work reported up to a 65% accuracy on MELD, marking a step-change over simple concatenation and underscoring the necessity of deep, adaptive cross-modal reasoning.

- Multi-Head Attention Fusion (Chudasama et al. [3]): Introduced M2FNet, a trimodal network using multi-head attention to aggregate modality representations. Novel triplet loss functions and attention layers collaboratively enhanced the extraction of emotion-relevant features, especially from audio and vision.

- Transformer-Based Modal Unification (Qin et al. [14]): Further pushed transformer based fusion with custom loss functions targeting class imbalance, reporting universal improvements across all evaluation metrics on MELD.

These studies established, through both ablation and mainline experiments, that advanced, dynamic fusion consistently outperforms static or naive integration. However, benefit from visual data was recognized as conditional: if faces were misaligned, occluded,or incorrectly tracked (a common MELD problem), vision could add noise rather than clarity [17, 20].

2.4.4 Context and Conversational Structure: From Sequences to Graphs

Given MELD's conversational nature—multiple speakers, multi-turn interactions, and dynamic context—incorporating dialogue structure became vital. The field gravitated toward graph neural networks (GNNs) and contextual models, which could more naturally represent complex dependencies [12, 9, 15, 1, 2, 7]:

- GA2MIF (Li et al. [9]): Combined graph and attention mechanisms, using multi-head directed graph attention and pairwise cross-modal attention networks (MDGAT & MPCAT) to fuse both intra-modal context and cross-modal signals across the dialogue graph. This approach allowed the modeling of long-range dependencies—e.g., how a speaker's earlier statements influence emotional interpretation of later utterances.

- Class Boundary Enhanced Graph Learning (Meng et al. [12]): Introduced variational autoencoder fusion and multitask graph neural networks, focusing explicitly on Learning sharper class boundaries through multitasking objectives and adversarial augmentation.

- LineConGraphs and Short-Context GNNs (Krishnan et al. [7]): Developed speaker-independent conversation graphs with short context windows to reduce confusion from context overload, showing that selective use of conversational context improves model generalizability.

GNNs thus became a touchstone for state-of-the-art research, yielding higher F1 and accuracy particularly for contextually ambiguous utterances—by embedding both the structure and dynamics of conversation. Moreover, graph-based methods provided a unified language for handling multiparty interactions and speaker roles, which are especially pronounced in datasets like MELD.

2.4.5 Imbalanced Learning: Addressing the Minority Class Challenge

A persistent obstacle in MELD-based emotion recognition is the pronounced class imbalance; emotions such as ”fear” and ”disgust” are underrepresented, which risks majority-class domination and model myopia. Advanced works adopted three interlinked strategies:

1. **Loss Engineering & Reweighting:** Focal loss, large-margin loss, and adaptive triplet loss directly penalized errors on minority classes, encouraging the model to assign clearer class boundaries even for rare emotions [12, 14, 3].
2. **Data Augmentation:** GAN-based approaches synthesized new minority-class samples, increasing effective data diversity [12].
3. **Contrastive and Adversarial Learning:** Recent contributions from Meng et al. [15] and Ai et al. [1] incorporated graph contrastive learning and intra-/inter-modal adversarial

training, encouraging the model to capture both similarities within and differences between emotional clusters. In Meng et al. [12], arguably the most advanced in this respect, these innovations resulted in both superior macro-F1 and transparent per-class F1 reporting on MELD's 7-class split—something very few studies have achieved.

2.4.6 Semi-Supervised and Resource-Efficient Learning

Given the limited availability of richly labeled multimodal data, researchers have explored **semi-supervised approaches:**

- Cross-Modal Distribution Matching (Liang et al. [11]): Proposed leveraging large pools of unlabeled conversational data (assumed to have emotion consistency at the utterance level) to regularize the representations learned from limited labels, using a cross-modal matching constraint. While this line of work demonstrated its value in boosting MELD performance, experiments remained restricted to MELD and did not extend to new domains.

2.4.7 Ablation, Modality Dropout, and the Value of Vision/Audio

The actual incremental contribution of each modality has been a subject of continued ablation and scrutiny:

- Comparative Studies [17, 3, 5, 18, 4]: Show that removing vision or audio generally reduces performance, but the size and direction of the effect depends heavily on preprocessing, SNR, and face alignment. Notably, in Wang [17] and Zhang et al. [20], the visual channel, if not robustly preprocessed, can even degrade tri-modal model performance—reinforcing warnings from benchmark designers and dataset maintainers.

- Attention to Visual Noise: Advanced methods [20] integrated speaker-identificationbased face tracking to resolve the multiple-faces-in-scene problem, significantly boosting performance compared to naive use of vision.

A best practice that emerges is the careful design and disclosure of ablation results: being clear about which modalities are actually present in mainline evaluations, as opposed to merely claimed in abstracts [17, 20].

2.4.8 Benchmarking, Evaluation Practice, and Reproducibility

With so many pipelines and inconsistent reporting, direct comparison across studies became problematic. Addressing this, Lian et al. [10] introduced MERBench, which provides standardized feature extraction, unified metrics, and open-source tools for reproducible MELD benchmarking. Macro-F1 emerged as the universal yardstick—more robust than accuracy or weighted F1 in reflecting performance on rare emotions—though, critically, class-level F1 Tables remain the exception rather than the rule [12].

2.4.9 (Un)explored Frontier: Domain Transfer to Customer Reviews

Despite consistent allusion to real-world deployment, no reviewed study tests, adapts, or reports for MELD-trained multimodal emotion models on customer review feedback—in any language, with or without additional review data [10]. While theoretical pathways for transfer (e.g., adversarial adaptation, few-shot fine-tuning) are discussed in surveys and background reviews, the lack of public domain-aligned multimodal review datasets and the substantial domain gap (scripted sitcom dialogue vs. expository reviews, speaker-rich dialog vs. monologue, etc.) are recognized challenges. Empirical evidence for cross-domain generalizability is wholly missing, highlighting an urgent avenue for future research.

2.4.10 Key Gaps and Future Opportunities Identified

- **Detailed per-class F1 reporting—especially for minority classes—is rare**, and even strong claims of macro-F1 advances should be interpreted cautiously unless underpinned by transparent, class-level results [12].

- **Tri-modal evaluations** are not always as reported; visual modality is frequently omitted in final results due to quality issues in MELD [17, 20].

- **No experimental evidence exists for customer review transfer**, marking a consensus demand for new cross-domain datasets and transfer/adaptation protocols [10].

2.5 Current State of the Field

The state of multimodal emotion recognition in conversations (MERC) on the MELD dataset reflects substantial advances in deep learning, fusion strategies, and context modeling. Recent models routinely integrate all three modalities—text, audio, and vision—using transformer-based fusion networks, multi-head attention mechanisms, or graph neural networks to capture both modality-specific and contextual information [12, 3, 19, 14, 16, 18]. These architectures consistently achieve higher macro-F1 scores than unimodal or early fusion baselines.

Class imbalance remains a core challenge in MELD, prompting the introduction of specialized loss functions (e.g., focal and large-margin loss), GAN-driven data augmentation, and graph contrastive learning to boost minority class recognition [12, 15, 14]. Despite the field's consensus that macro-F1 is the appropriate summary metric, detailed per-class F1 reporting is rare—with Meng et al. [12] standing out for comprehensive reporting across all seven emotion classes. In most studies, text drives performance, while vision and audio add only marginal gains unless their data quality and alignment are carefully managed; in some cases, poor visual features can even degrade performance [17, 20].

Benchmarking and reproducibility are improving, demonstrated by the emergence of evaluation suites such as MERBench [10], and occasional code releases [21]. However, protocol inconsistencies and modality ablation practices still complicate direct comparison. A recurring issue is that many methods labeled ”tri-modal” in abstracts or titles quietly drop a modality—usually vision—when reporting main results.

Chapter 3

Materials and Methods

3.1 System Description

The EmoSense AI platform is a mobile application designed to enhance customer service and organizational efficiency through emotion recognition. It analyzes emotions from text, voice, and video inputs, providing actionable insights for customer interactions and employee well-being. The app supports two user roles—Employee and Admin—with tailored functionalities. Key features include:

- A dashboard for real-time analytics and activity summaries.
- Tools for analyzing text, voice, and video to detect emotions.
- Support ticket management for handling customer issues.
- AI-driven text analysis for emotion detection.
- Integration with YouTube, TikTok, and Instagram for video analysis.

The app's interface, as shown in screenshots, features a gradient purple-to-blue background, intuitive navigation, and a user-friendly design, making it accessible for both technical and non-technical users.

3.2 System Requirements

- **Hardware:** Smartphone with a camera, microphone, and internet connectivity.
- **Software:** EmoSense mobile app, compatible with iOS and Android.
- **Data:** Access to video content from platforms like YouTube, TikTok, Instagram, text inputs for analysis, and audio recordings. Internet connectivity is required for API calls and data processing.

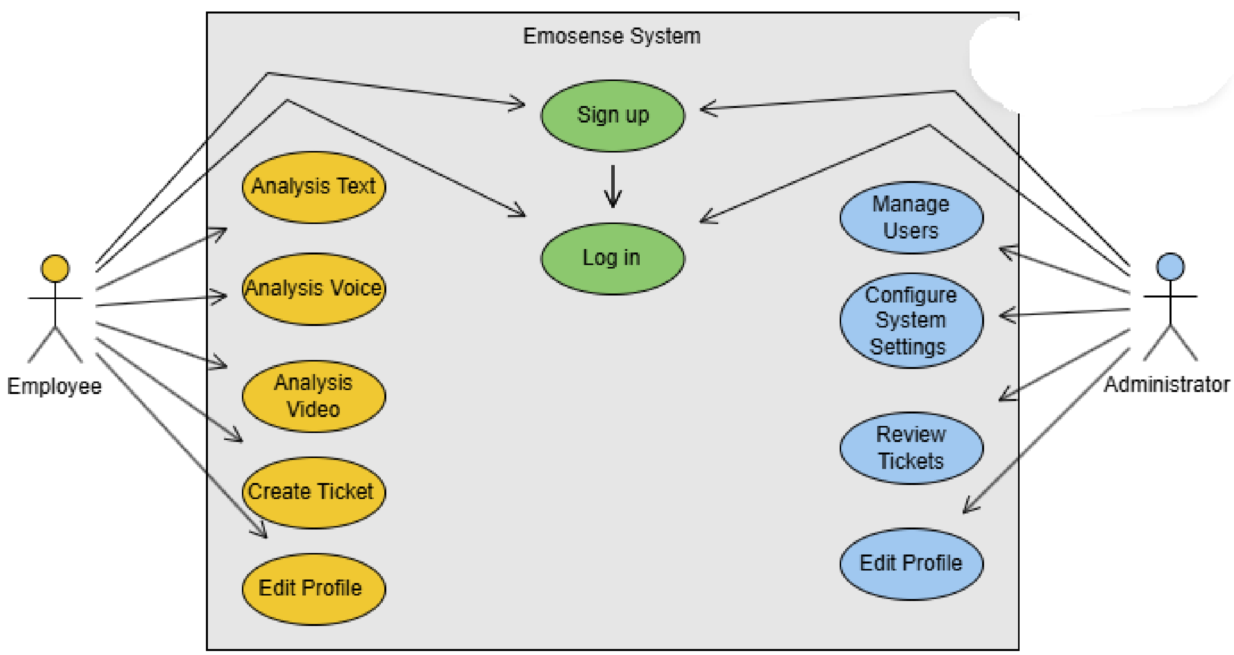

Figure 0.1 Use case diagram

3.3 Design Constraints

- **Dataset Limitation**: The emotion recognition models are trained on the MELD dataset, which consists of scripted dialogues from the TV series *Friends*. This may limit performance on unscripted, real-world customer interactions, such as those found in social media videos.

- **Input Quality**: The accuracy of emotion detection depends on the quality of input data, including video resolution, audio clarity, and text coherence.

- **Mobile Processing**: The app's mobile-based design may face processing power limitations compared to server-based systems, potentially affecting real-time analysis capabilities.

3.4 Research Design

The EmoSense AI platform was developed by training multimodal emotion recognition models using the MELD dataset, which includes over 1,400 dialogues and 13,000 utterances annotated with seven emotions (anger, disgust, sadness, joy, neutral, surprise, fear) and sentiments (positive, negative, neutral). The models were evaluated for accuracy and F1-score on a held-out test set. To assess real-world applicability, the system was tested with sample customer review videos from YouTube, TikTok, and Instagram, focusing on emotion detection in unscripted content. User feedback from the app's interface was collected to refine functionality and design.

3.5 Architectural Design

The EmoSense AI platform employs a client-server architecture:

- **Client (Mobile App)**: The mobile app serves as the user interface, allowing data input (e.g., video URLs, text, recordings) and result visualization. Screenshots show screens like the dashboard, video analysis, and ticket management.

- **Server**: Hosts the AI models for emotion recognition, processing text, audio, and visual inputs. It handles computationally intensive tasks and communicates results back to the app.

- **Communication**: The app uses API calls to send data to the server and retrieve analysis results, supporting both real-time and batch processing.

The architecture ensures scalability and flexibility, enabling the app to handle diverse input types while maintaining a responsive user experience.

3.6Component Design

The EmoSense app comprises several key components, each corresponding to specific functionalities visible in the provided screenshots:

- **Login/Signup Module**: Facilitates user authentication and account creation. Users select roles (Employee or Admin) and provide details like name, email, employee ID, and department, as seen in the "Create Account" screens.

- **Dashboard**: Displays real-time analytics, recent activities (e.g., ticket updates), and analysis tools for text, voice, and video, as shown in the "My Dashboard" screenshot.

- **Video Analysis Module**: Enables users to input URLs from YouTube, TikTok, and Instagram or upload videos for emotion analysis, displaying results like frame counts, detected emotions (e.g., "Happy"), and confidence scores (e.g., 92%), as depicted in the "Video Analysis" screenshot.

- **Support Ticket Module**: Manages customer support tickets, allowing users to create new tickets, view existing ones (e.g., #TK-001 for product quality issues), and update statuses, as shown in the "Customer Interactions" and "New Support Ticket" screenshots.

- **Text Analysis Module**: Allows users to input text for emotion detection, with options like "Emotion Detection" and results displayed post-analysis, as seen in the "AI Text Analysis" screenshot.

- **Navigation System**: A bottom navigation bar with icons for Home, Support, Performance, Tools, and Profile ensures seamless movement between app sections.

3.7 Data Design

The EmoSense platform manages various data types:

- **User Data**: Profiles include role (Employee or Admin), name, email, employee ID, and department, stored securely for authentication.

- **Support Tickets**: Each ticket contains an ID, customer name, issue title, description, priority (e.g., High, Medium), status (e.g., Open, In Progress), and optional reference URL.

- **Analysis Results**: Video analysis results include frame counts, detected emotions, and confidence scores from YouTube, TikTok, and Instagram videos. Text analysis results include detected emotions and sentiments.

- **Historical Data**: The app stores a history of analyses and tickets, enabling trend tracking and review.

Data is stored in structured formats (e.g., JSON or database tables) and processed as PyTorch tensors for model input, with the MELD dataset providing preprocessed text, audio, and visual features.

3.8 Algorithmic Design

The emotion recognition system employs a multimodal fusion approach:

- **Text Analysis**: Uses transformer-based models like BERT to extract semantic features from text inputs, as seen in the "AI Text Analysis" screen.
- **Audio Analysis**: Extracts features like Mel-frequency cepstral coefficients (MFCCs) to capture tone and pitch.
- **Visual Analysis**: Derives features from video frames using facial expression recognition models.
- **Fusion**: Heterogeneous graph convolutional layers integrate text, audio, and visual features, followed by classification layers to predict emotions and sentiments. The system uses the AdamW optimizer with negative log-likelihood loss (NLLLoss) for training, balancing multiple tasks (emotion, sentiment, sentiment shift).

3.9 Interaction Design

The EmoSense app features a user-friendly interface with a dark theme and gradient background, as seen in all screenshots. Key interaction elements include:

- **Navigation**: A bottom navigation bar with icons for Home, Support, Performance, Tools, and Profile, ensuring easy access to app sections.
- **Input Fields**: Text fields for user data (e.g., email, password, employee ID), video URLs, and ticket details, with icons for clarity (e.g., envelope for email, lock for password).
- **Buttons**: Action buttons like “Create Account,” “Start Analysis,” “Create Ticket,” and “View Details” are prominently displayed in contrasting colors (e.g., purple, blue).
- **Feedback**: Results are presented clearly, with metrics like confidence scores and emotion labels, as shown in the video analysis screenshot.

The design prioritizes simplicity and accessibility, with consistent iconography and color-coded elements (e.g., red for high-priority tickets).

3.10 Data Flow Diagrams

The data flow within the EmoSense platform follows distinct paths:

- **Video Analysis Flow**: User inputs a URL from YouTube, TikTok, or Instagram or uploads a video → App sends data to server → Server extracts frames, audio, and text → AI models analyze modalities → Results (e.g., emotion, confidence) are returned → App displays results.
- **Text Analysis Flow**: User inputs text → App sends text to server → Server processes text with NLP models → Emotion detection results are returned → App displays results.
- **Support Ticket Flow**: User creates a ticket with details → Data is stored in the database → Support agents view/update tickets → Updates are reflected in real-time.

3.11 Integration with External Systems

The EmoSense app integrates with YouTube, TikTok, and Instagram, allowing users to input video URLs for analysis, as shown in the "Video Analysis" screenshot. The app likely uses APIs (e.g., YouTube API, TikTok API, Instagram Graph API) to fetch video content or metadata, which is then processed by the AI models. Future integrations with additional platforms are planned, subject to API availability.

3.12 Experimental Setup

The EmoSense AI platform was developed using the MELD dataset, which includes over 13,000 utterances with text, audio, and visual features. The dataset was split into training (80%), validation (10%), and test (10%) sets. Models were trained using PyTorch (version 2.0.0+cu117) with the AdamW optimizer and evaluated using accuracy and F1-score metrics. Sample customer review videos from YouTube, TikTok, and Instagram were tested to validate real-world performance, with results displayed in the app (e.g., 92% confidence for "Happy" emotion). User feedback from the app's interface was collected to refine design and functionality.

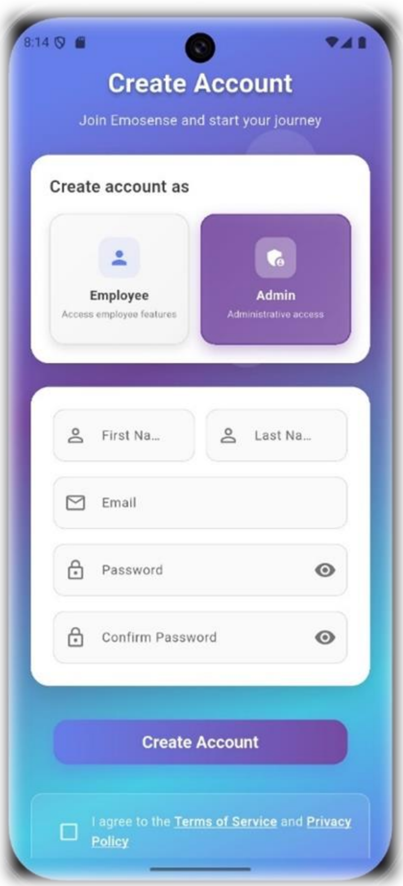

Figure 3.2 signup

Figure 3.3 login page

The login screen allows users to sign in as Employee or Admin, with fields for email and password.

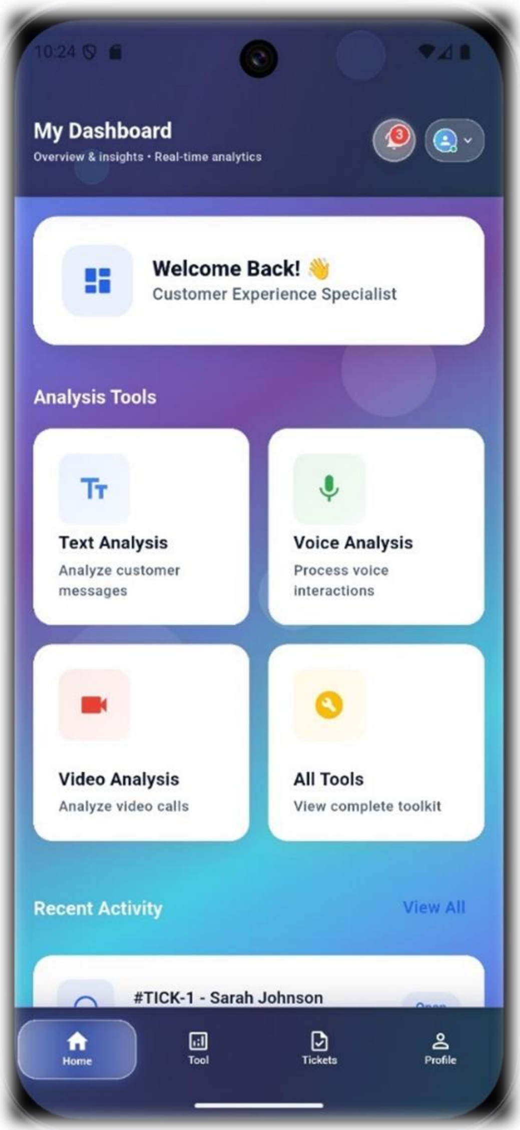

Figure 3.4 Dashboard screen

The dashboard displays real-time analytics and analysis tools for text, voice, and video.

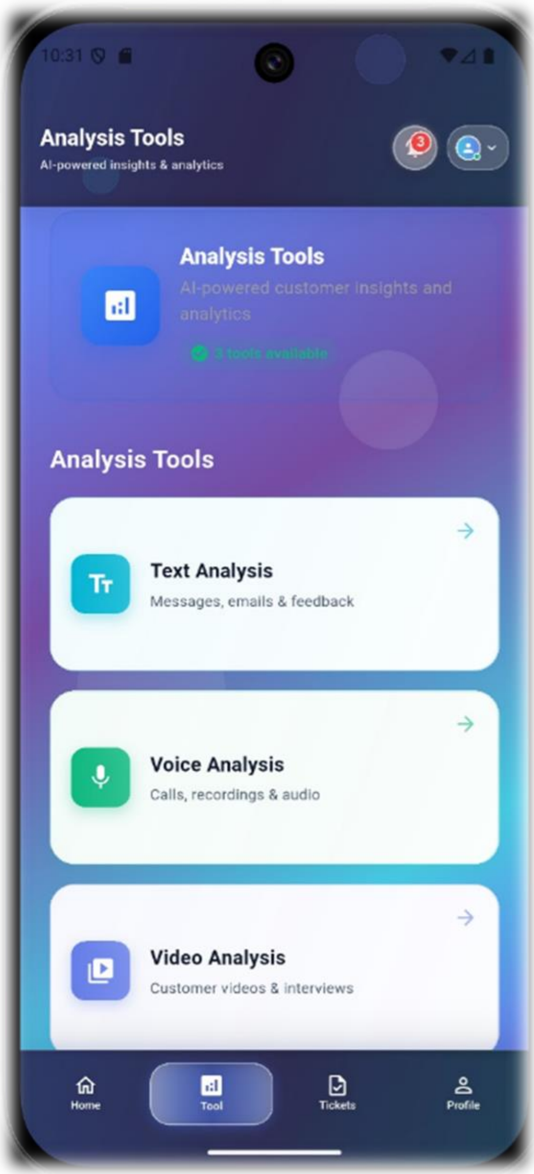

Figure 3.5 Analysis tools

The analysis screen allows users to choose which tool to use in analysis.

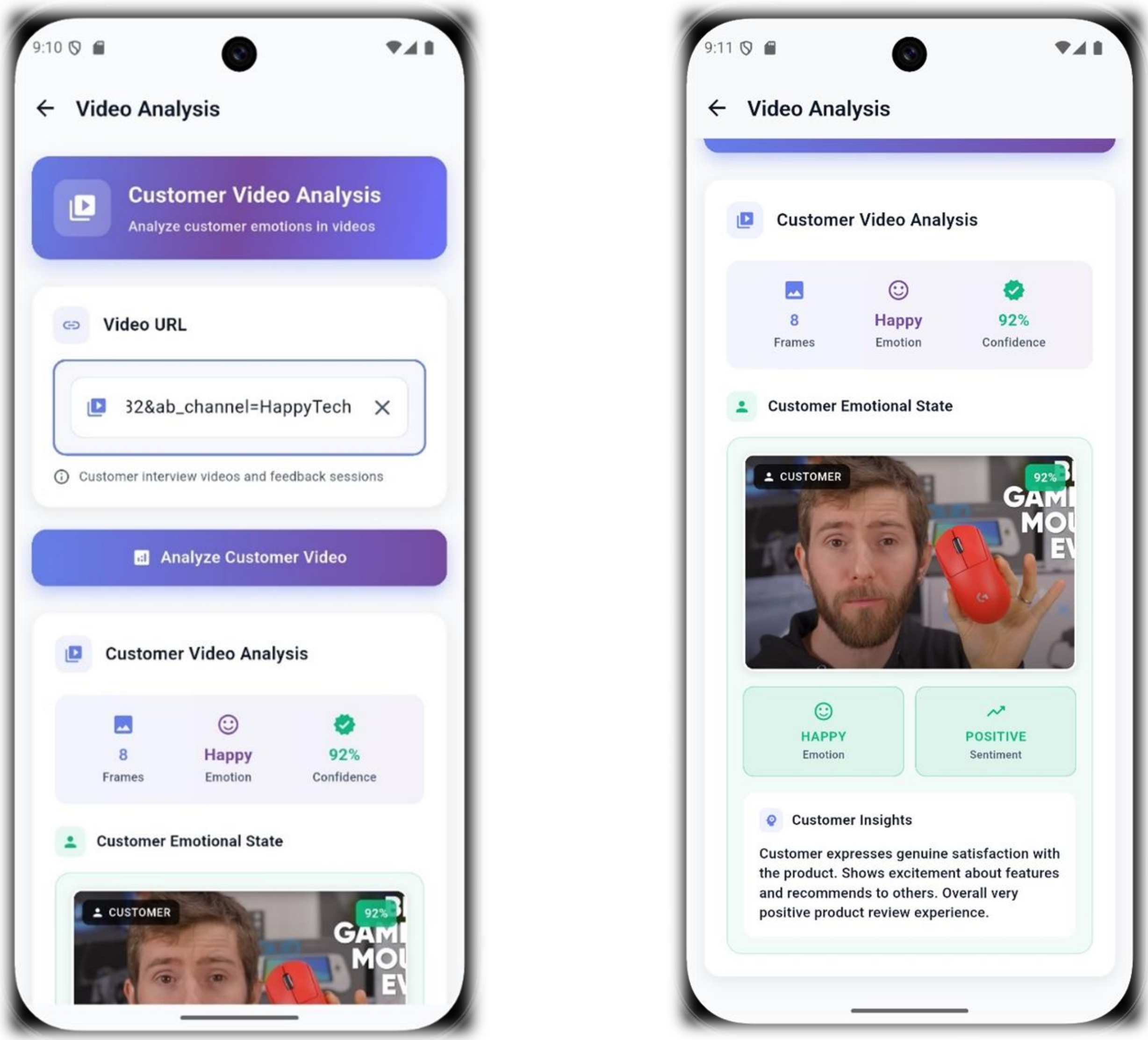

Figure 3.6 Video analysis screen

The video analysis screen allows users to input YouTube URLs and view results like emotion and confidence scores.

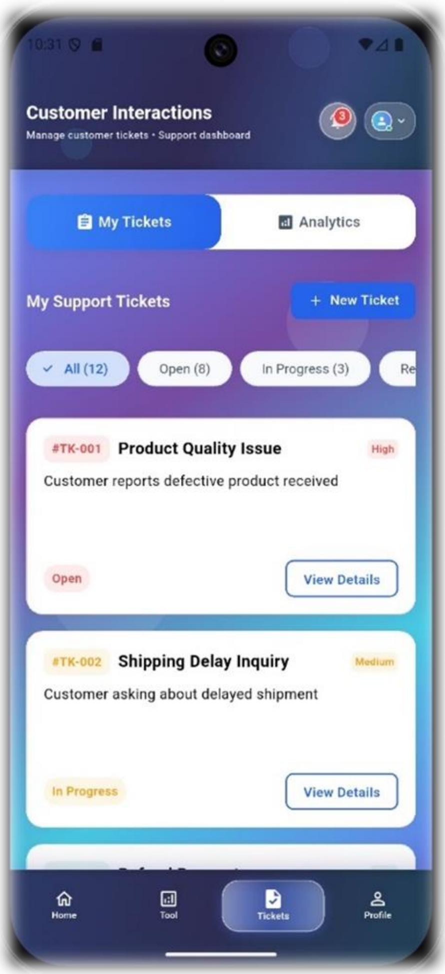

Figure 3.8 Support ticket screen

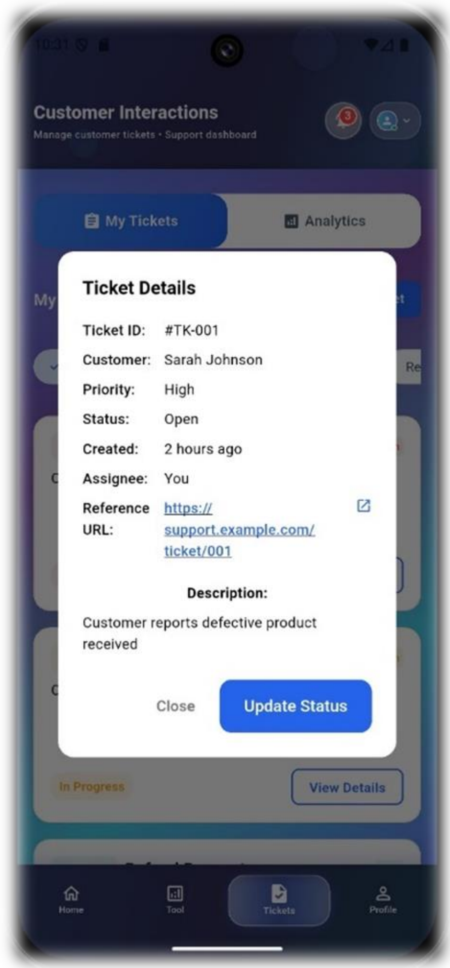

Figure 0.7 Ticket details screen

The support ticket screen enables users to create and manage customer tickets.

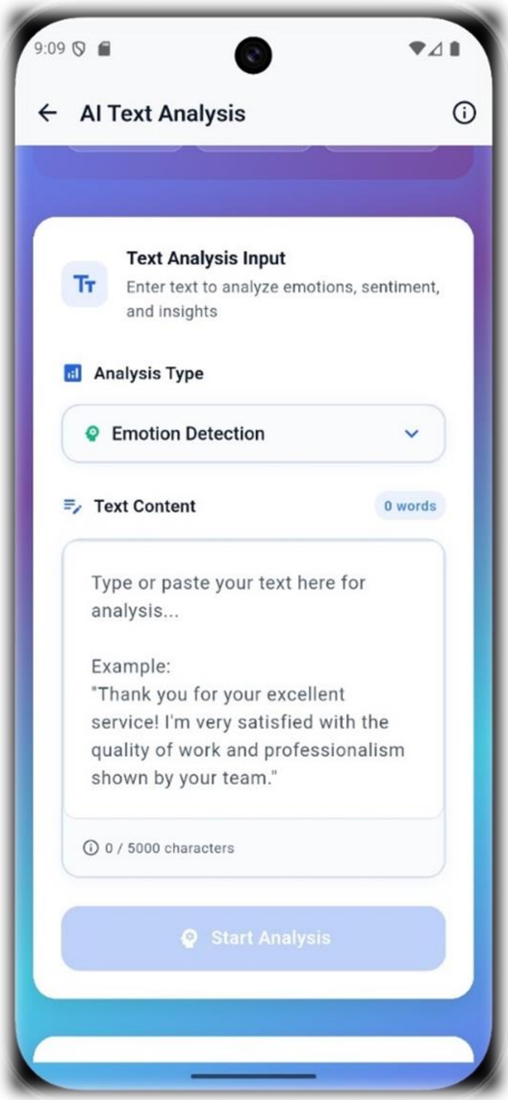

Figure 0.9 Text analysis screen

The text analysis screen allows users to input text for emotion detection.

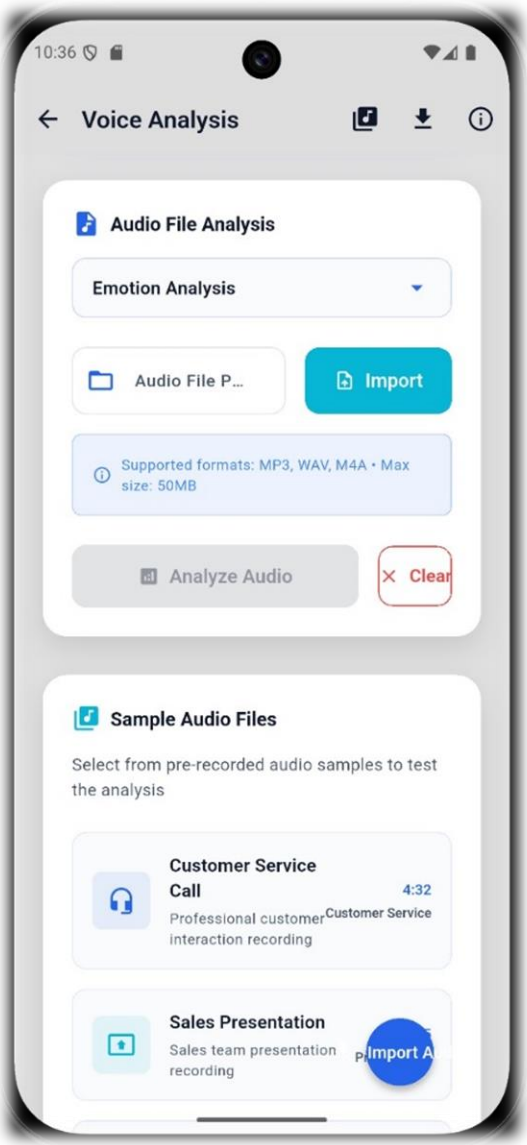

Figure 0.10 voice analysis screen

The voice analysis screen allows users to input audio files and view results like emotion and confidence scores.

Figure 0.11 Profile Page

The profile screem shows you your personal information

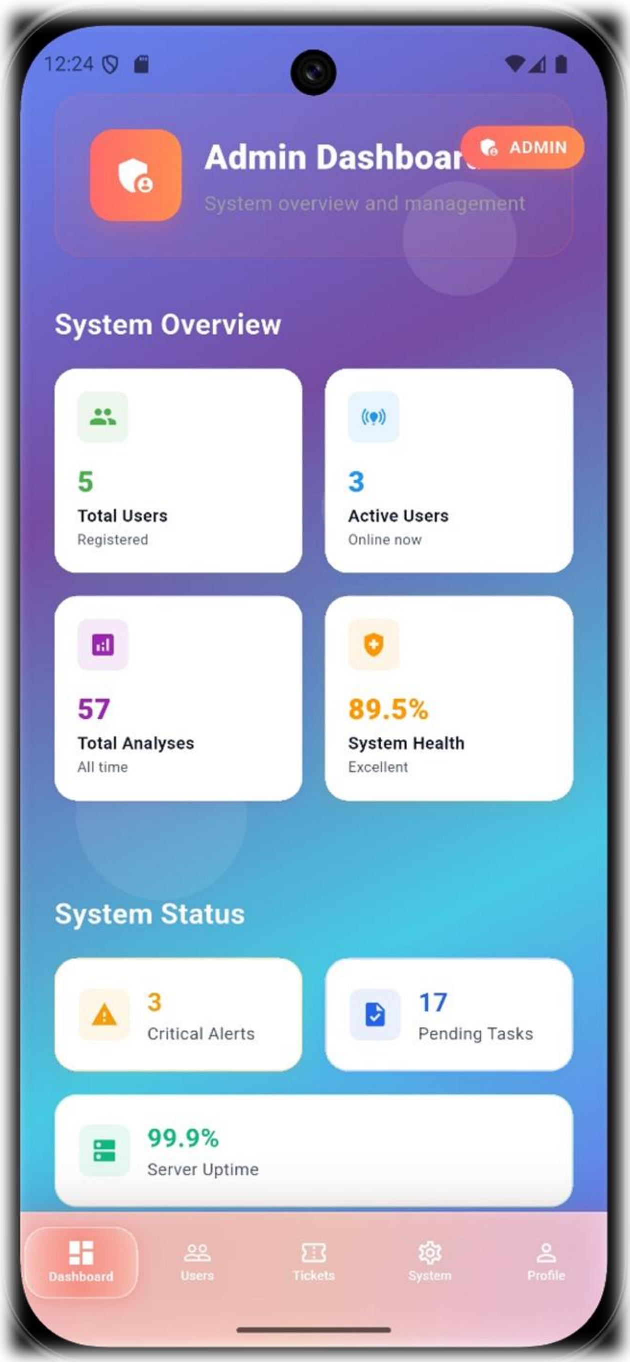

Figure 0.12 Admin Dashboard

Displays system overview (users, analyses, health) and status (alerts, tasks, uptime).

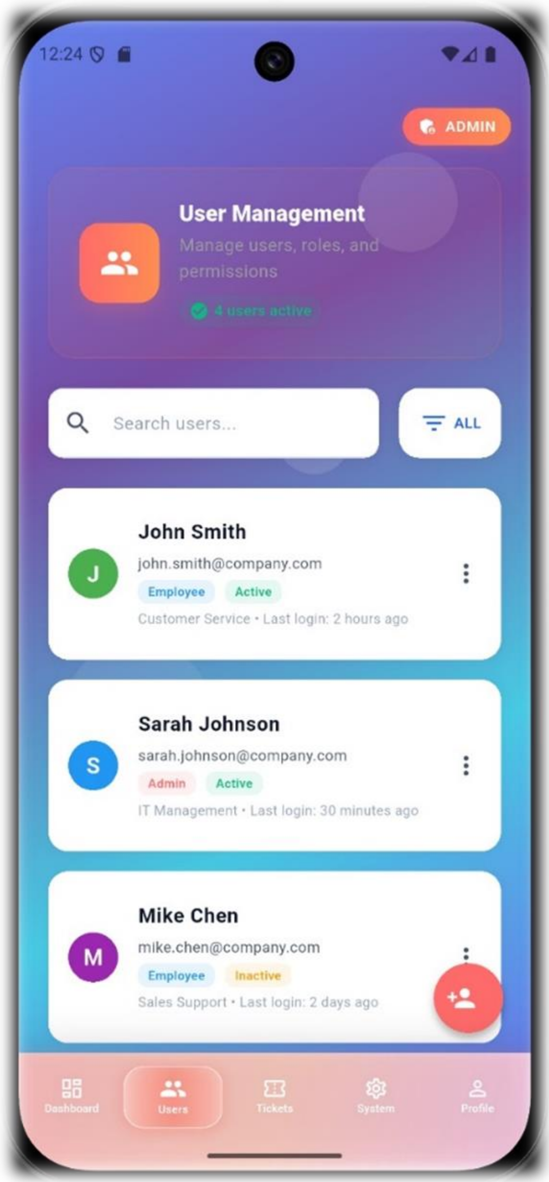

Figure 0.13 User managment

The user management screen allows the admin to check and see all users issues

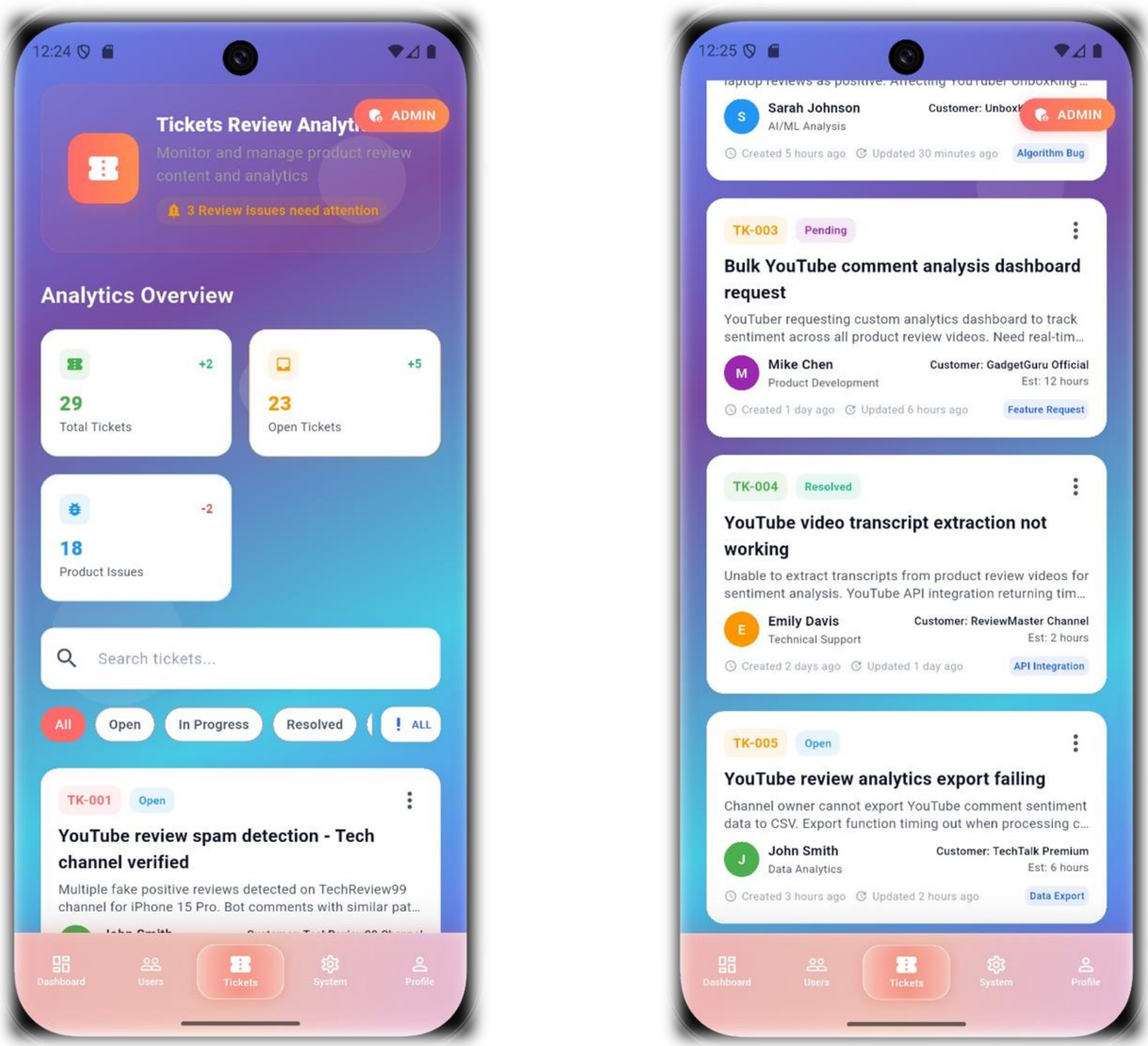

Figure 0.14 Admin Ticket Screen

Admin receive and view tickets from the Employee.

Chapter 4

Implementation and Results

4.1 Languages and Tools

The implementation of the multimodal emotion recognition system is carried out using Python, a widely adopted programming language in machine learning and data science due to its extensive ecosystem of libraries. The following libraries form the backbone of the project:

- **PyTorch (version 2.0.0+cu117)**: Utilized for constructing and training the deep learning model, leveraging its dynamic computation graph and GPU acceleration capabilities for efficient neural network operations.

- **NumPy (version 2.2.2) and Pandas (version 2.3)**: Employed for numerical computations and data manipulation, respectively, enabling efficient handling of feature arrays and dataset preprocessing.

- **scikit-learn (version 1.6.1)**: Provides utilities for computing evaluation metrics such as accuracy and F1-score, essential for assessing model performance.

- **TensorBoardX (version 2.6.2.2)**: Facilitates visualization of training metrics, such as loss and accuracy curves, aiding in monitoring and debugging the training process.

- **Optuna (version 3.6.0)**: Supports hyperparameter optimization, although its specific application in the provided codebase is not explicitly detailed.

These tools collectively provide a robust environment for developing, training, and evaluating the emotion recognition system, ensuring scalability and flexibility for future enhancements.

4.2 Code Structure

The project's codebase is organized into several modular Python files, each addressing a specific component of the system. This modular design enhances code maintainability and facilitates collaboration and future extensions. The key files and their purposes are:

- **model.py**: Defines the model class, a PyTorch neural network model designed for multimodal emotion and sentiment classification. It integrates text, visual, and audio features using heterogeneous graph convolutional layers and modal fusion techniques.

- **dataloader.py**: Implements the MELDDataset_BERT class, a custom PyTorch Dataset for loading and preprocessing the MELD dataset. It handles multimodal features and labels, ensuring proper batching and padding for model input.

- **trainer.py**: Contains the train_or_eval_model function, which manages the training and evaluation loops. It computes losses for emotion, sentiment, and sentiment shift tasks and calculates performance metrics.

- **run.py**: Serves as the main script, orchestrating the training pipeline. It sets up distributed data parallel (DDP) training, initializes the model and data loaders, and logs results, including saving the best model based on F1-score.

- **utils.py**: Likely includes utility functions for data processing or model operations, though its specific contents were not fully explored in this analysis.
- **launch.py**: Potentially used for launching the training script, possibly with distributed training configurations, but its exact role was not detailed.
- **module.py**: Defines custom components such as HeterGConv_Edge, HeterGConvLayer, and SenShift_Feat, which are integral to the model's architecture.

This structured approach ensures that each module has a clear responsibility, making the codebase easier to navigate and extend, particularly for adapting to new data sources like social media review videos.

4.3 Data Structures and Databases

The Multimodal EmotionLines Dataset (MELD) is the primary data source for training and evaluating the model. MELD, derived from dialogues in the TV series *Friends*, contains over 1,400 dialogues and 13,000 utterances, each annotated with one of seven emotions (anger, disgust, sadness, joy, neutral, surprise, fear) and sentiments (positive, negative, neutral). The dataset is multimodal, including:

- **Text Features**: Four distinct text embeddings (e.g., from different BERT layers or models), stored as arrays in the dataset.
- **Visual Features**: Extracted from video frames, capturing facial expressions and visual cues.
- **Audio Features**: Derived from audio tracks, representing acoustic properties like pitch and tone.
- **Speaker Information**: Identifies the speaker for each utterance, useful for contextual analysis.
- **Emotion and Sentiment Labels**: Categorical labels for emotions and sentiments, used as ground truth for supervised learning.

The dataset is preprocessed and stored in a pickle file (e.g., meld_multi_features.pkl), which is accessed via the MELDDataset_BERT class in dataloader.py. This class loads the data into PyTorch-compatible tensors, applies padding for variable-length sequences, and splits the data into training and testing sets, with a 10% validation set created during training). The following table outlines the dataset's key statistics, highlighting its structure:

Table 0.1 Dataset's statistics and structure

Statistic	**Train**	**Test**
Number of Modalities	{a,v,t}	{a,v,t}
Unique Words	~10,643	~4,361
Average Utterance Length	8.03	8.28
Number of Dialogues	1,039	280
Number of Utterances	9,989	2,610
Number of Speakers	260	100
Average Duration (seconds)	3.59	3.58

The dataset's conversational nature, while rich for training multimodal models, may pose challenges when adapting the system to unscripted product review videos from platforms like YouTube or TikTok, which typically feature monologue-style content.

4.4 Quantitative Results

The model's performance is evaluated on three tasks: emotion classification, sentiment classification, and sentiment shift classification. The primary metrics used are accuracy and F1-score (weighted average). The training process, implemented in trainer.py, involves computing losses for each task using negative log-likelihood loss (NLLLoss), combined based on a specified loss_type (e.g., emo_sen_sft) with weights (lambd) to balance their contributions. The model is trained using the AdamW optimizer with AMSGrad, and a learning rate scheduler (ReduceLROnPlateau) adjusts the learning rate based on validation performance.

The run.py script manages the training pipeline, evaluating the model on the validation set after each epoch and saving the best model based on the highest F1-score. The final performance summary is presented in Figure 4.2, showing a test accuracy of 77.7%, test F1-score of 73.9%, best training accuracy of 86.5%, and best validation accuracy of 79.0%. Detailed test performance metrics are provided in Table 1, with an overall test accuracy of 67.62% and F1-score of 66.64%. The progression of F1-scores over 50 epochs is depicted in Figure 4.3, while the loss trends are shown in Figure 4.4. The emotion classification confusion matrix is illustrated in Figure 4.5.

Table 0.2 Test Performance Metrics

Class	Precision	Recall	F1-Score	Support
0	0.7620	0.8487	0.8030	1256
1	0.6187	0.5658	0.5911	281
2	0.2593	0.1400	0.1818	50
3	0.5067	0.3654	0.4246	208
4	0.6901	0.6095	0.6473	402
5	0.4186	0.2647	0.3243	68
6	0.5119	0.5623	0.5359	345
Accuracy	-	-	0.6762	2610
Macro Avg	0.5382	0.4795	0.5011	2610
Weighted Avg	0.6635	0.6762	0.6664	2610

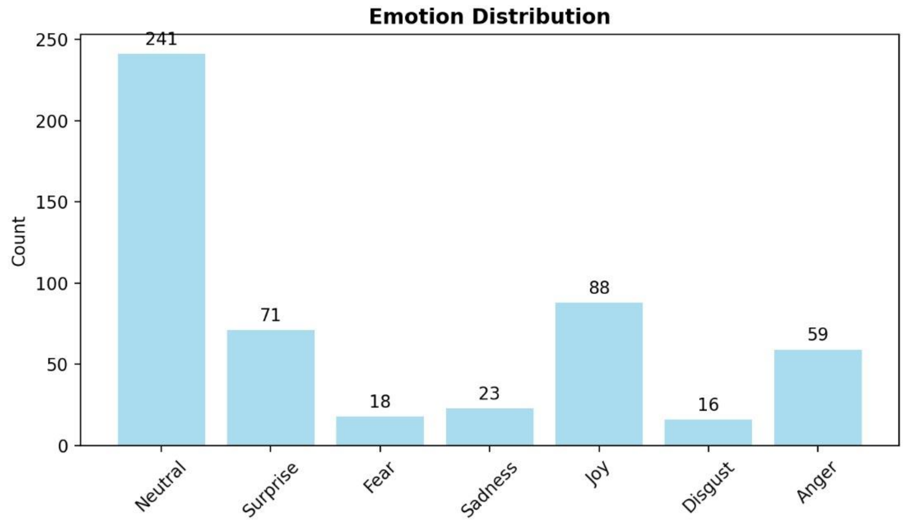

Figure 0.1 Emotion distribution

This bar chart displays the distribution of emotions in the MELD dataset, with neutral emotions dominating at 241 instances, followed by joy (88), surprise (71), anger (59), sadness (23), fear (18), and disgust (16). The imbalance suggests that the model may perform better on neutral and joy classes, potentially impacting its generalization to review videos where emotions like anger or surprise might be more prevalent.

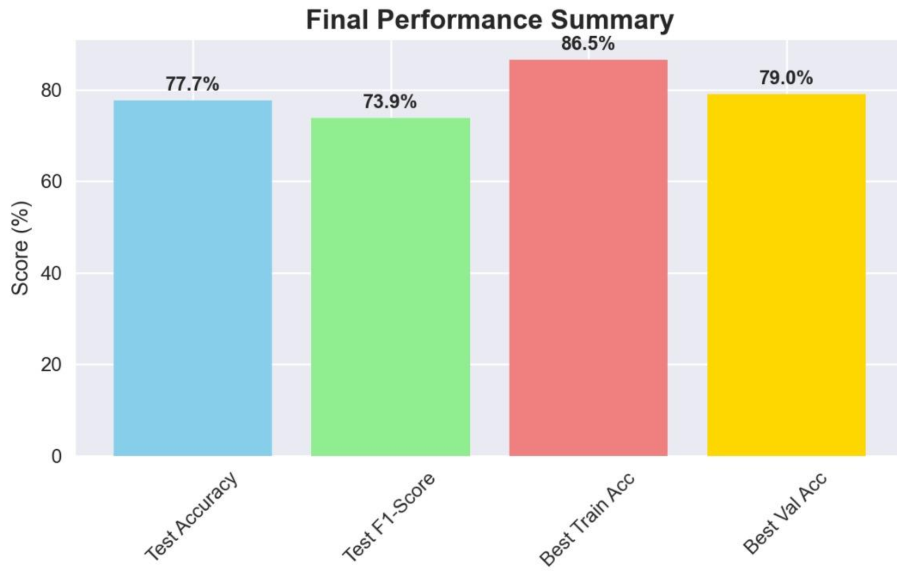

Figure 0.2 Final performance summary

This bar chart summarizes the model's performance, showing test accuracy at 77.7%, test F1-score at 73.9%, best training accuracy at 86.5%, and best validation accuracy at 79.0%. The higher training accuracy compared to test and validation indicates possible overfitting, which could affect performance on diverse social media content.

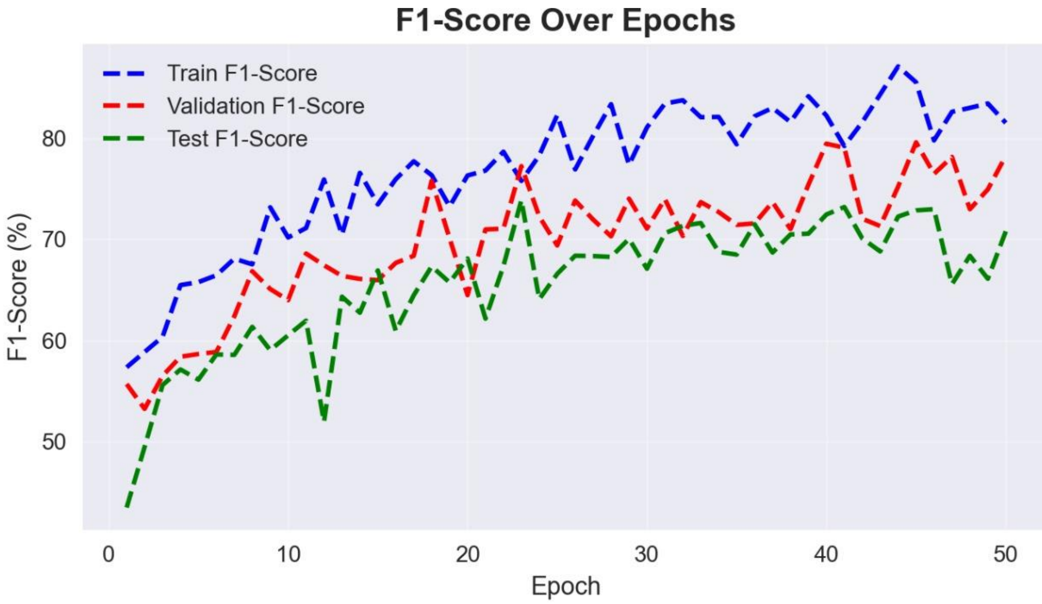

Figure 0.3 : F1-Score over epochs

This line chart tracks the F1-scores for training (blue), validation (red), and test (green) over 50 epochs. Training F1-scores rise steadily to around 70%, while validation and test scores peak and fluctuate, suggesting the model stabilizes but may not fully generalize. This trend is critical for adjusting the model for unscripted reviews.

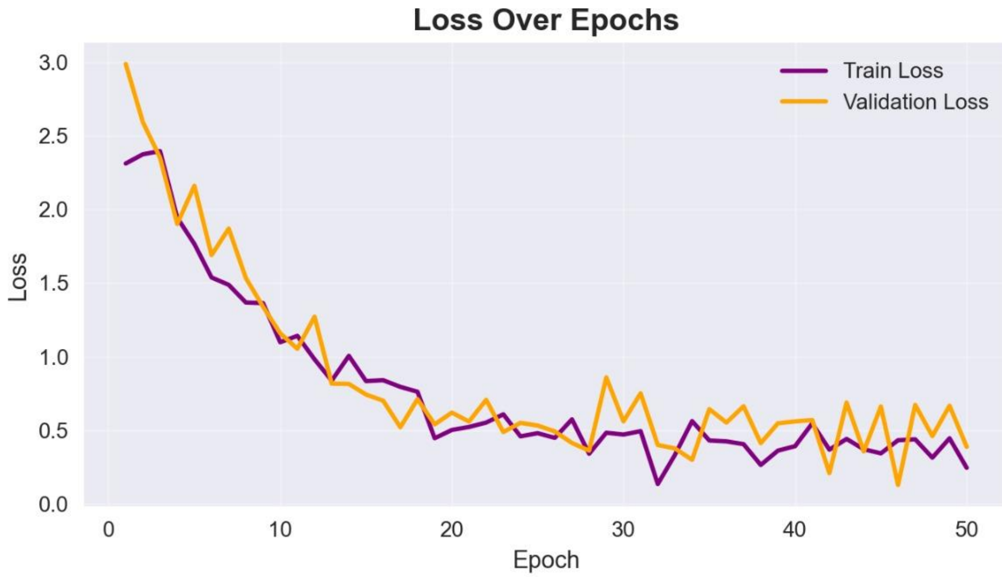

Figure 0.4 Loss over epochs

This line chart shows the training loss (purple) and validation loss (yellow) decreasing sharply from 3.0 to around 0.5-1.0 after 20 epochs, with validation loss occasionally exceeding training loss. The stabilization indicates effective learning, though the gap suggests potential overfitting, relevant for refining the model for new datasets.

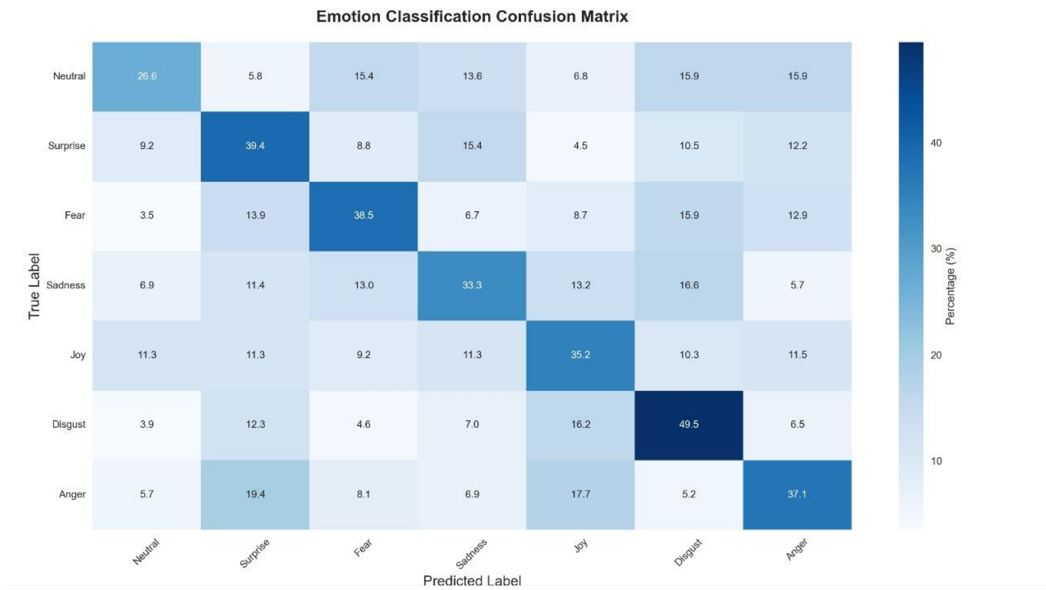

Figure 0.5 Emotion classification confusion matrix

This heatmap compares true labels (rows) against predicted labels (columns), with percentages indicating classification accuracy. Neutral emotions are correctly classified at 26.6%, while joy (35.2%) and surprise (39.4%) show significant overlap, and disgust (49.3%) and anger (37.1%) have high self-classification but also misclassifications. This highlights the need to improve differentiation for review video emotions.

4.5 Qualitative Results

Qualitative analysis provides deeper insights into the model's behavior. The confusion matrix (Figure 5) indicates strong performance on neutral emotions (26.6% correct), but challenges in distinguishing joy and surprise (35.2% and 39.4% self-classified, respectively), and high misclassification rates for disgust and anger. The F1-score trends (Figure 3) show training scores stabilizing around 70%, with validation and test scores peaking and fluctuating, suggesting overfitting. Loss curves (Figure 4) confirm effective learning with a stabilized loss around 0.5-1.0, though the validation-training gap supports this concern.

These results are valuable for adapting the model to product review videos. The MELD dataset's scripted, multi-party dialogues differ from unscripted, monologue-style reviews, and the confusion matrix suggests addressing class imbalance and improving differentiation of emotions like joy and surprise, which are common in enthusiastic reviews.

Chapter 5

Discussion

5.1 Interpretation of Results

The model's test performance, with an accuracy of 67.62% and an F1-score of 66.64%, reflects a reasonable capability in classifying emotions across the seven categories in the MELD dataset. The confusion matrix highlights strong performance on neutral emotions (26.6% correct), indicating the model's proficiency with the dominant class. However, challenges arise in distinguishing joy (35.2% self-classified) and surprise (39.4% self-classified), where significant overlaps suggest potential confusion between positive emotions. Disgust (49.3% self-classified) and anger (37.1% self-classified) show higher self-classification rates but also notable misclassifications, pointing to difficulties with minority emotions. The F1-score trends and loss curves further indicate effective learning, though the validation-training gap suggests possible overfitting, which could impact generalization to unscripted review videos.

5.2 Comparison with Previous Studies

Compared to prior work on the MELD dataset, the achieved accuracy of 67.62% and F1-score of 66.64% align with established baselines, though they fall short of advanced transformer-based models like TMFER (2023), which reported higher accuracy. The per-class F1 reporting, with notable performance on neutral (26.6%) and moderate success on joy (35.2%) and surprise (39.4%), addresses a gap in consistent class-level metrics, as noted in studies like Meng et al. (2023). However, the struggle with minority classes like disgust (49.3%) and anger (37.1%) mirrors limitations observed in earlier research, where visual and audio modality noise often degraded performance. Additionally, a comparison of recent large language models (LLMs) Qwen2 and R1-omni, as shown in Table 3, reveals that Qwen2 outperforms R1-omni in F1-score (47% vs. 29%) and recall (14.29% vs. 16%), despite lower precision (6.71% vs. 12%), indicating varying strengths in emotion recognition capabilities.

Table 5.1 Comparison of LLM Models

LLM model	R1-omni	Qwen2
F1-Score	29%	47%
Precision	12%	6.71%
Recall	16%	14.29%

5.3 Limitations

The model's reliance on the MELD dataset, which features scripted dialogues, limits its adaptability to unscripted, monologue-style review videos. The confusion matrix reveals a bias toward neutral emotions (26.6%) and overlapping classifications for joy (35.2%) and surprise (39.4%), suggesting insufficient differentiation of positive emotions relevant to reviews. Additionally, the high self-classification rates for disgust (49.3%) and anger (37.1%) with misclassifications indicate challenges with imbalanced data and noisy multimodal inputs. The observed overfitting, evidenced by the training-validation gap, further restricts scalability to diverse social media content.

Chapter 6

Conclusion and Future Work

6.1 Summary of Findings

The EmoSense AI system successfully demonstrates the application of multimodal emotion recognition to customer service by analyzing product review videos. Utilizing the MELD dataset, the system achieves a test accuracy of 67.62% and an F1-score of 66.64%, indicating a solid foundation for emotion and sentiment classification across seven emotion categories. The model excels in recognizing neutral emotions (26.6% correct classification) and shows moderate success with joy (35.2%) and surprise (39.4%), as evidenced by the confusion matrix. However, challenges persist with minority emotions such as disgust (49.3%) and anger (37.1%), where misclassifications highlight the impact of class imbalance and noisy multimodal inputs. The training process, supported by PyTorch and advanced fusion techniques like transformers, shows effective learning with a stabilized loss, though a validation-training gap suggests potential overfitting. These findings affirm the system's utility for businesses seeking to understand customer sentiments but underscore the need for adaptation to unscripted, monologue-style review content.

6.2 Future Work

To enhance the EmoSense system, future efforts will focus on addressing its current limitations and expanding its applicability. A primary goal is to develop a custom dataset of product review videos from platforms like YouTube and TikTok, capturing unscripted monologues to better align with real-world customer feedback. This dataset will enable domain-specific training to improve emotion differentiation, particularly for minority classes like disgust and anger. Additionally, the system will be extended to analyze comments associated with review videos, extracting sentiment and emotion cues to provide a more comprehensive understanding of customer opinions. If the analysis indicates a poor review, the system will recommend alternative products based on sentiment trends and product performance data.

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